ALTERNATIVE DATA SCORING MODELS

Alternative data scoring models use non-traditional information sources to assess creditworthiness, especially for the unbanked or thin-file populations common in Malawi.

Traditional credit bureaus rely on loan repayment history, but alternative data leverages everyday digital and behavioral footprints.

WHY ALTERNATIVE DATA MATTERS FOR MALAWI

1. Malawi has low formal credit penetration, a large informal sector, and many people without banking history
2. Mobile money (e.g., via Airtel Money or TNM Mpamba), airtime purchases, and agricultural activities offer rich signals of financial behavior.
3. TransUnion Malawi actively promotes alternative data like mobile money transactions, airtime purchases, and utility payments to assess reliability for underserved groups.

   This approach supports financial inclusion while helping lenders reduce risk

KEY TYPES OF ALTERNATIVE DATA RELEVANT TO MALAWI

Practical categories with examples and predictive value:

1. Telco & Mobile Money Data----Airtime top-up frequency/consistency, data usage patterns, mobile money transaction history (volume, frequency, inflows/outflows), call/SMS patterns, SIM age.
   Strong predictor of cash flow stability and repayment ability.
2. Utility & Bill payments------Electricity (ESCOM), water or pay-as-you go history
   Shows payment discipline.
3. Psychometric & Behavioral Data-----Gamified quizzes assessing traits like conscientiousness, risk aversion, or honesty
4. Transaction & Cash-flow Data------Merchant payments, gig economy or small business inflows (e.g., via mobile)
5. Social & Network Data--------(used cautiously) social connections repayment behavior or group lending history (e.g., village savings and loans groups)
6. Other Emerging Sources------Agricultural data (e.g., crop yields, Market prices via digital platforms), geolocation/mobility patterns, or e-commerce activity.

COMMON SCORING MODEL APPROACHES

1. Logistic Regression on Telco Data----Simple, interpretable models using mobile network operator (MNO) data effective for unbanked populations in Africa.

2. Machine Learning / AI models-----Handle complex interactions and large datasets. Studies show significant AUC improvements (e.g., up to 0.79) when combining alternative with traditional data.
3. Hybrid models-----Blend traditional credit bureau data with alternative sources for better accuracy on both thin-file and existing customers.
4. Psychometric + Digital footprint----Combine quizzes with mobile data for character + capacity assessment
5. Cash-flow Based Scoring-----Real-time analysis of inflows/outflows, ideal for MSMEs and informal traders

   Alternative data often improves risk prediction by 17-126% in various segments and helps score previously unscorable individuals